

KB-003-Hardware-Fault-Diagnosis

KB-003 — Hardware Fault vs. Vehicle-Side Issue Diagnosis

Category: Hardware Diagnosis
Call Volume: ~18% of contacts
Risk: Misattribution wastes technician dispatches and frustrates drivers
Primary Remote Actions: Status check, hard reset, technician dispatch

Why This Matters

Hardware failure is one of the most misdiagnosed categories in EV charging support. Drivers frequently assume the charging station is broken when the fault actually originates in the vehicle — particularly at the charging port, the battery management system (BMS), or the vehicle’s onboard communication controller. Dispatching a technician to a healthy charger is expensive and delays resolution for the driver. Equally, dismissing a genuine charger fault as a vehicle problem leaves a broken unit in the field damaging all future users.

This article provides a systematic framework to distinguish charger-side from vehicle-side faults using phone-based questioning.

Background: Where Faults Live

Charger-Side Hardware Faults

Component	What can fail	Typical symptom
Power module / rectifier	Can’t convert AC to DC	Session starts, no power flow; screen may show output fault
Connector / cable	Physical damage, bent pins, worn latch Processing failure	Connector won’t lock; intermittent charging

Component	What can fail	Typical symptom
Control board / firmware		Screen frozen, unresponsive, random resets
Payment terminal	Reader failure	Card declined across all cards; screen error
Network modem	No OCPP connectivity	Station shows offline on backend; can't authorize
Ground fault detector	Nuisance tripping or real fault	E005 / GFCI error; do not reset — safety issue
Thermal management system	Overheating protection	Throttles or shuts down in hot weather
Display / UI	Screen failure	Black screen, partial display; may not affect charging

Vehicle-Side Faults

Component	What can fail	Typical symptom
CCS / CHAdeMO inlet	Physical damage, bent pins, debris, corrosion	Won't connect or intermittent loss
Onboard charger (AC)	Internal fault	AC charging fails; DC fine, or vice versa
Battery Management System (BMS)	Communication dropout, calibration error	Session starts then drops; car shows warning
Pilot signal circuit	Can't negotiate power level with EVSE	Handshake fails; error on charger screen
Contactors / relay	Won't close to allow charge	Session authorized but no power flows into battery
Charging limit setting	Soft limit set by driver or manufacturer	Session ends early at a specific SoC

Diagnostic Framework

Phase 1 — Pattern Questions (Ask Before Anything Else)

These five questions narrow the fault location before any remote action is taken.

Q1: Does this happen at only this station, or at other stations too?

- Only this station → likely charger fault. Prioritize reset and tech dispatch.
- Multiple stations → likely vehicle fault. Do not dispatch a tech.

- Only DC fast chargers → narrow to DC inlet or BMS; AC charging may work.
- Only AC chargers → onboard charger suspect; DC may work.

Q2: Have you successfully charged at this same station before?

- Yes, recently → something changed. Could be charger degradation or a vehicle change.
- Never tried here before → no baseline; could be either side.
- Yes, years ago but not recently → charger hardware may have degraded.

Q3: What does the charger screen show?

Screen display	Likely fault location
Blank / black screen	Charger control board or power supply
“Communication error” or “Vehicle not responding”	Vehicle-side (BMS or pilot circuit)
“Connector fault” or “Latch error”	Physical connector issue — could be either
“Ground fault” (E005)	Ground fault — safety issue, do not reset
“Authorized” but no power flow	Could be either; check contactor
“Charging” but car not receiving power	Likely vehicle-side (BMS, contactor)
No error, just stuck on “Starting...”	Backend authorization timeout or network issue

Q4: What does the vehicle display show?

- No warning lights → vehicle presenting as ready; charger fault more likely
- “Charging fault” or specific error code → vehicle-side likely
- “Check charging system” → vehicle-side; recommend dealer visit
- Lightning bolt or plug icon with X → vehicle’s onboard charger may be flagging rejection

Q5: Is the vehicle charging port visibly clean and undamaged?

Ask the driver to inspect: - Any debris, moisture, or foreign objects in the inlet - Bent or damaged pins (they should not touch these) - Discoloration or burn marks (safety flag — escalate) - Corrosion (visible rust or green deposits)

Phase 2 — Remote Action Based on Phase 1

If charger fault is suspected:

1. Check real-time station status in the backend
2. If online: attempt soft reset, then hard reset
3. If offline or faulted: confirm to driver, skip reset, flag for dispatch

4. Log the failure pattern with all Phase 1 answers for the technician

If vehicle fault is suspected:

1. Do not dispatch a technician to the charger — it is unnecessary
 2. Script: > “Based on what you’re describing — especially that this is happening at multiple different stations — the issue is most likely with your vehicle’s charging system rather than any specific charger. I know that’s not the answer you were hoping for. I’d recommend contacting your vehicle manufacturer or an EV-certified service center. They can run a diagnostic on your charging port and battery management system.”
 3. Log: vehicle make, model, year, fault description, charger types attempted (AC/DC), any error codes from vehicle display
 4. Offer to document the call for the driver to share with their dealership
-

Section A — Specific Hardware Fault Scenarios

Scenario 1: Black or Frozen Screen

“Is the rest of the station on — are any lights active, or is the whole unit dark?”

- Whole unit dark: power supply failure or tripped breaker. Hard reset via backend if possible; if no response, dispatch.
- Screen off but charger has indicator lights: display failure only — charger may still function via app or RFID start.
- Screen frozen on startup logo: firmware crash. Hard reset. If persistent: dispatch.

Scenario 2: Charger Shows “Charging” But Car Isn’t Receiving Power

This is a subtle fault — the charger thinks it’s working but the vehicle isn’t accepting energy.

Checklist: 1. Confirm the vehicle’s in-car display shows the charging indicator active 2. Ask: has the state of charge moved in the last 5+ minutes? 3. Ask: is there a charge limit set in the vehicle? (may have already hit the cap)

If car SoC isn’t moving and no vehicle-side limit is set: - The contactor (relay that closes to allow current flow) may not be closing, either on the charger side or vehicle side - Hard reset the charger; if unresolved, advise vehicle service

Scenario 3: Error Code “Ground Fault” (E005 or GFCI)

Do not reset this station.

“I need to flag this station as a safety issue — a ground fault error means there’s a potential electrical problem at the unit. Please unplug your cable and step away from the station. I’m going to mark it for a technician right now.”

Log as a safety-priority dispatch. Do not allow any driver to use this station until the technician clears it.

Scenario 4: Cable / Connector Damage

Ask the driver to look at the connector end (the end that goes into the car) and describe what they see.

- **Bent pins:** Do not attempt to insert — this can damage the vehicle inlet. Dispatch a technician. Offer the driver an alternative station.
- **Burn marks or discoloration:** Safety flag. Remove from service immediately. See KB-009.
- **Debris in socket:** Ask driver to not insert the connector. A technician or the driver (carefully, with the charger unplugged from the car) can remove debris with dry compressed air.
- **Cracked or broken housing:** Remove from service. Dispatch.

Scenario 5: Intermittent Charging (Keeps Starting and Stopping)

This pattern — charging for a few minutes, stopping, restarting — is most commonly caused by:

1. **Loose connector fit** — re-seat the cable; may be a worn latch on the charger or the vehicle inlet
2. **Thermal protection cycling** — charger or vehicle overheating in hot weather; normal behavior but frustrating
3. **Grid instability at the site** — voltage fluctuations trip the charger’s protection; dispatch an electrician via the tech team
4. **BMS communication dropout** — vehicle-side; recommend vehicle service

“That start-stop pattern can happen for a few reasons. Can I ask — is it a hot day, and has the car been sitting in the sun? [If yes] Some chargers and vehicles throttle or cycle in high heat to protect the battery — frustrating, but it’s a safety feature. [If no] Let me try a reset and see if that stabilizes it.”

Section B — Distinguishing Charger vs. Vehicle at Scale

If the same station generates repeated calls from different drivers: - **Fault is at the charger** — dispatch for hardware inspection regardless of remote reset results

If the same driver calls repeatedly across different stations: - **Fault is at the vehicle** — document thoroughly and direct to manufacturer; do not keep dispatching techs

Pattern data to log every call: - Station ID - Vehicle make, model, year - Connector type used (CCS1, CCS2, CHAdeMO, NACS) - AC or DC session - Error code (charger screen + vehicle display) - Whether other stations succeeded or failed for this driver

Technician Dispatch Criteria

Dispatch a technician when:

Condition	Priority
Ground fault error (E005)	SAFETY — immediate
Burn marks or damage to connector	SAFETY — immediate
Hard reset fails, station shows online	High
Connector physically stuck (see KB-008)	High
Same station fails for 3+ different drivers in one day	High
Screen is blank and hard reset has no effect	Medium
Intermittent charging that persists after reset	Medium

Agent Reminders

- Never diagnose a vehicle fault with certainty over the phone — say “it may be worth checking” not “your car is broken.”
- Always log vehicle details (make, model, year, connector type) even when the fault is charger-side — it helps technicians prepare.
- A ground fault (E005) is a hard stop. Do not reset. Do not let the driver retry. Flag and escalate immediately.
- If the driver describes sparks, burning smell, or visible damage: go directly to KB-009 Safety Protocols.